

Project Completion Plan

The Second Brain: Grounded Personal Memory and Reflection Platform

Course: OENG1185 / COSC2503 Capstone Project Part B / Programming Project 2

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1. Updated Project Description

The Second Brain is an authenticated personal intelligence platform that converts diary entries, calendar context, attachments, and summaries into searchable memory. The current MVP is not only a journal: it stores user-authored context, indexes it asynchronously into memory chunks, retrieves relevant evidence using hybrid vector and lexical search, and returns source-grounded answers with citations, confidence, timing, and token analytics.

The scope has been refined since the original proposal. Instead of expanding equally into every Google service and every file type, Capstone B now prioritises a measurable core path: Diary -> Indexing Worker -> Memory Retrieval -> Cited Answer -> Summary/Reflection. This adjustment responds to supervisor and mentor feedback that the system must prove trust, retrieval accuracy, and latency before adding more integrations.

Scope item	Current decision	Justification
Core diary memory	In scope and implemented	Main value path; mood/tags were added so emotional reflection can be evaluated.
Google Calendar	In scope and partly live	OAuth, token storage, event sync, diary linking, and memory indexing are implemented; Google test-user verification remains an operational dependency.
Attachment ingestion	In scope for MVP testing	Upload, Supabase Storage, text extraction path, worker indexing, and attachment citations exist; PDF/image/DOCX extraction still needs more acceptance testing.
Gmail and Google Contacts	Deferred/future work	Schema and planning exist, but the team decided not to expand beyond Calendar until the cited memory loop is stable.
AI optimisation	High priority	Focus is on retrieval filters, reranking, grounded fallback, cache control, and measurable evaluation metrics.

2. Completed Work and Current Status

The project has moved from a concept into a working modular monorepo with a Next.js frontend, NestJS API, PostgreSQL/Supabase data layer, pgvector memory storage, and a Node worker for background jobs.

Area	Completed evidence from code	Remaining gap
Diary and Timeline	Authenticated diary CRUD, mood selector, tag input, timeline display, edit/delete, signed attachment URLs.	Final UI polish and mobile walkthrough testing.
AI Memory	Gemini embeddings, semantic chunking, hybrid retrieval, date/source filters, reranking, citation validation, debug trace, fast/deep/auto modes, fallback for quota or invalid JSON.	Run benchmark dataset and record Recall@5, citation precision, no-answer accuracy, and p95 latency.
Calendar	Server-side OAuth callback, token refresh update, encrypted token support, event upsert, sync endpoint, diary-event linking, calendar event indexing jobs.	Google consent screen test users must be maintained; more live-account regression tests required.
Attachments	Upload endpoint supports text/PDF/image/DOCX MIME types, Supabase Storage, Attachment rows, text extraction for plain text, worker extraction/indexing for	Complete controlled PDF/image/DOCX acceptance tests and document extraction accuracy.

	non-text files, citation-capable chunks.	
Summary/Reflection	Daily, weekly, monthly, yearly summary pipeline; summaries are queued for memory indexing and shown in Summary UI.	Tune prompts and evaluate summary usefulness with user tasks.
Observability/QA	Search UI exposes confidence, answer mode, pipeline timing, token usage, retrieved chunks, cache/fallback status; seed scripts and worker drain support repeatable demo data.	API Jest suite has one module-resolution failure; worker build config and web build must be rechecked outside sandbox constraints.

Feedback response: the report and implementation now emphasise measurable evidence. The AI answer layer rejects unsupported names/dates, uses citation checks before returning model answers, and falls back to extractive evidence rather than inventing answers. The team also added seed data and explicit demo readiness checks so testing is repeatable.

3. Planned Deliverables and Dates

The remaining plan is organised around five delivery weeks. The first three weeks match the team's current test plan: real attachment ingestion, Google Calendar integration, and AI optimisation.

Week / dates	Owner focus	Deliverables	Acceptance evidence
W1: 21-27 Jul	Attachment ingestion	Upload real PDF/image/DOCX/text files; verify Storage, extraction, outbox jobs, memory chunks, and attachment citations.	At least 3 file types indexed; search answer cites attachment source; no dead-letter jobs.
W2: 28 Jul-3 Aug	Google Calendar	Retest OAuth with approved Google test users; sync real events; refresh token; link diary entries to events; index Calendar memories.	Connected status true; events in DB; linked diary count > 0; Calendar query returns cited event.
W3: 4-10 Aug	AI optimisation	Run benchmark questions, refine filters/reranking/fallbacks, separate retrieval latency from generation latency, disable stale cache in tests.	Recall@5, citation precision, no-answer accuracy and p95 retrieval latency recorded.
W4: 11-17 Aug	Reflection and readiness	Improve summary quality, demo readiness dashboard, seed data, error/loading states, and requeue controls.	Create -> Search -> Summary flow passes with realistic data.
W5: 18-24 Aug	Final demo and report	User testing, bug fixing, deployment rehearsal, final written evidence and showcase script.	Demo script completed twice; user testing results summarised; high risks closed or scoped.

4. Quantifiable Success Metrics

Metric	Target	Feature measured	Evaluation method
Retrieval Recall@5	>= 80%	AI Memory retrieval	Use seeded benchmark questions with known expected source chunks.
Citation precision	>= 85%	Grounded answer/citation UI	Manually verify whether each

			citation supports the claim shown.
No-answer accuracy	>= 80%	Hallucination control	Ask unsupported questions and check that the system refuses instead of inventing.
p95 retrieval latency	<= 500 ms excluding Gemini generation	Memory search backend	Record embed/retrieve/generate timing separately from Search analytics.
Indexing reliability	0 dead-letter jobs after drain; >= 95% succeeded	Worker and outbox pipeline	Run seed/drain scripts and inspect readiness/outbox status.
Calendar sync success	>= 90% of test events synced and searchable	Google Calendar integration	Compare Google Calendar test events with `calendar_events` and search citations.
User task completion	>= 80% completion across 5 core tasks	Frontend/product UX	Test users complete create diary, attach file, connect Calendar, ask Search, read Summary.

5. Required Resources

Resource	Why it is required
Supabase Postgres + pgvector + Storage	Stores authenticated users, diary entries, Calendar events, attachments, summaries, memory chunks, and vectors.
Gemini API quota	Required for embeddings, chunk generation, answer generation, summary generation, and attachment text extraction for non-text files.
Google Cloud OAuth test users	Required because the app is in Testing mode; unapproved accounts receive 403 access_denied.
Vercel/Render or equivalent hosting	Needed for stable frontend/API demo and OAuth callback URLs.
Weekly mentor/team integration time	Needed to close cross-module risks: API contracts, UI binding, worker indexing, and demo evidence.

6. Risk Management Plan

Risk	Priority	Mitigation / workaround
LLM hallucination or weak citation grounding	High	Keep strict RAG prompt, citation validation, unsupported-name/date checks, low-confidence fallback, and visible source snippets.
Gemini quota/rate limits block live indexing or answering	High	Throttle worker jobs, cache completed answers, prepare seeded memory chunks, and maintain backup API key or lower-cost model for indexing.
Google OAuth app blocks unapproved testers	High	Add all team/test accounts to Google OAuth test users; verify redirect URI before demo; document reconnect process.
Async indexing jobs stuck or dead-lettered	High	Use outbox status/readiness checks, requeue controls, drain script, and demo checklist

		before each rehearsal.
Attachment extraction quality varies by file type	Medium	Accept plain text as baseline, test PDF/image/DOCX separately, and mark advanced OCR/DOCX extraction as controlled MVP scope.
Frontend/backend contract mismatch	Medium	Freeze DTOs for Diary, Search, Summary, Calendar; run API client tests; use mock data only as fallback demo mode.
Documentation overclaims implementation	Medium	Keep report aligned with code evidence and explicitly separate completed, partial, and future work.

Priority rule: if a high-risk item remains unresolved in any week, the team will stabilise the core MVP path first and move optional scope such as Gmail, Google Contacts, advanced OCR, and data export into future work.

Appendix A. Current Verification Snapshot

Checked on 19 July 2026 against the local repository: AI package tests pass 40/40; worker tests pass 2/2; web tests pass 4/4. AI and API builds pass. API Jest has one suite blocked by a module-resolution issue, and web production build failed in the sandbox because Turbopack could not bind a local process; both are tracked as QA/configuration gaps before final submission.